Study Guide Question Set: Module 1

Subnational Climate-Migration Dynamics & Active Retrieval Companion | Petrova et al. (2026)

Instructional Design Note: This study guide operates as an *active retrieval scaffold*. Answering these open-ended prompts without pre-supplied answers drives goal-directed processing and leverages the testing effect to strengthen long-term conceptual retention.

Part 1: Foundational Scope & Methodology (Direct Retrieval)

Question 1 (The Macro-Micro Paradox):

• Why do national-level migration statistics obscure localized climate mobility patterns?

• What specific subnational administrative unit (Admin-2) did Petrova et al. (2026) analyze, across how many countries/territories, and over what timeframe?

Question 2 (Calculating Subnational Net Migration):

• How are annual subnational net migration figures derived using gridded **WorldPop** data and natural population change (births minus deaths)?

• What is the critical distinction between *net migration* and total *in-/out-migration volume*?

Part 2: Hazard Frameworks & Mechanistic Comparisons (Conceptual Analysis)

Question 3 (Population-Weighted Exposure Framework):

• Why did the researchers utilize a population-weighted intensity-exposure framework via ISIMIP climate models rather than relying solely on physical hazard boundaries?

• How does a drought in an unpopulated area differ in exposure metrics from a drought in a densely populated agricultural district?

Question 4 (Rapid-Onset vs. Slow-Onset Dynamics):

• Compare rapid-onset events (floods, cyclones) with slow-onset hazards (droughts) regarding the speed and timing with which their impacts show up in subnational net migration.

• Why do droughts typically display no statistically significant immediate (same-year) response in migration data?

Part 3: Critical Synthesis & Policy Application (Generative Reasoning)

Question 5 (Hidden Mobility Dimensions & Trapped Populations):
• Environmental stress does not always manifest as immediate outward displacement. How can severe climate shocks lead to *suppressed in-migration* or *resource-constrained immobility* ('trapped populations')?
• How do 4 consecutive years of severe flooding erode household resources needed to relocate?

Question 6 (Policy & Disaster Adaptation Planning):
• If a regional climate adaptation agency relies exclusively on national totals or single-year disaster reports, what key long-term migration trends will they miss?
• How does district-level monitoring improve targeted infrastructure, economic, and humanitarian planning?

Module 1 Core Reference Matrix

Concept / Parameter	Study Metric / Finding	Pedagogical / Analytical Significance
Subnational Unit	Admin-2 (districts / counties)	Eliminates national-level averaging; captures internal population shifts.
Global Sample	216 countries & territories (2000–2019)	Provides a 20-year multi-decadal panel of global subnational mobility.
Rapid-Onset Hazards	Floods & Tropical Cyclones	Triggers multi-year shifts; immediate impact on local receiving capacities.
Slow-Onset Hazards	Droughts & Crop Failures	Delayed response; requires multi-year compounding to alter mobility.